Task 5 — Ensemble Learning

Breast Cancer Wisconsin · Logistic Regression vs Random Forest vs Gradient Boosting · Hyperparameter Tuning with GridSearchCV

Author: Gundu Jeevesh Laahiri · **Internship:** Maincrafts Technology — AI & ML · **Dataset:** sklearn.datasets.load_breast_cancer

1. Project Objective

The objective of this task is to build a complete classification pipeline that compares a Logistic Regression baseline with two ensemble methods (Random Forest and Gradient Boosting) and then tune the Random Forest using **GridSearchCV**. The comparison exposes how ensemble capacity and hyperparameter optimisation translate — or fail to translate — into test-set performance on a real medical screening dataset. Ensemble learning is a core technique in production machine-learning systems because aggregating many weak or correlated learners is one of the most reliable ways to reduce variance; this experiment shows both the strengths and the limits of that idea.

2. Dataset and Preparation

The dataset is the **Breast Cancer Wisconsin (Diagnostic)** dataset bundled with scikit-learn. It contains **569 observations** with **30 numeric features** derived from fine-needle-aspirate image analysis and a binary target. The scikit-learn mapping is **0 = malignant** and **1 = benign**, verified programmatically before any model is trained. The class distribution is 212 malignant (37.3 %) and 357 benign (62.7 %) — moderately imbalanced in favour of the benign class. There are no missing values and no duplicate rows.

An **80/20 stratified train-test split** with random_state=42 was created *before* any preprocessing was fit, producing 455 training and 114 test observations with the class ratio preserved in both splits. The test set was held out and never used during model selection or tuning. Logistic Regression was wrapped with StandardScaler in a scikit-learn Pipeline so that scaling is re-fit on each cross-validation fold without leakage. Tree-based models were not scaled.

Because this is a medical-screening example, **malignant (class 0) is treated as the positive class** whenever precision, recall or F1 is reported, and the GridSearchCV scorer is make_scorer(f1_score, pos_label=0). Accuracy and ROC-AUC are label-symmetric. A column labelled “Malignant Recall” is included in the comparison table so the clinically important number is unambiguous.

3. Models Implemented

Four models were trained on the same training set and evaluated on the same held-out test set, using identical metric definitions:

Logistic Regression (baseline). StandardScaler → LogisticRegression(max_iter=5000, random_state=42) wrapped in a Pipeline. Serves as a strong but simple linear baseline against which the ensembles are judged.

Random Forest (default). RandomForestClassifier(random_state=42). An ensemble of 100 bagged decision trees trained on bootstrap samples with random feature subsets, which reduces variance relative to a single tree.

Gradient Boosting. GradientBoostingClassifier(random_state=42). Sequentially fits shallow trees to the residuals of the current ensemble; typically more accurate than a single learner but slower to train and more sensitive to hyperparameters.

Random Forest (optimized). Random Forest refit with the best hyperparameters chosen by GridSearchCV over 36 configurations (see §4).

4. Evaluation Method

Each model was evaluated with the same six metrics: **accuracy** (overall), **precision**, **recall** and **F1-score** computed with pos_label=0 (malignant), **ROC-AUC** using the malignant-class probability, and a 2×2 **confusion matrix** with malignant as the positive row/column. Fit time was recorded for a rough view of computational cost.

For the Random Forest search, GridSearchCV was configured with **5-fold stratified cross-validation**, n_jobs=-1 and the **malignant-F1** scorer. The grid covered n_estimators ∈ {100, 200, 400}, max_depth ∈ {None, 5, 10}, min_samples_split ∈ {2, 5}, min_samples_leaf ∈ {1, 2} — 36 configurations × 5 folds = 180 fits. Only the training set was used; the test set was reserved for the final comparison.

5. Results

Model	Accuracy	Prec.	Recall	F1	ROC-AUC	Mal. Rec.	Fit (s)
LogisticRegression	0.9825	0.9762	0.9762	0.9762	0.9954	0.9762	0.006
RandomForest (default)	0.9561	0.9512	0.9286	0.9398	0.9937	0.9286	0.180
GradientBoosting	0.9561	0.9744	0.9048	0.9383	0.9907	0.9048	0.369
RandomForest (optimized)	0.9561	0.9512	0.9286	0.9398	0.9937	0.9286	0.164

*Table 1. Held-out test-set performance of all four models. Precision, Recall and F1 are reported for the **malignant** class (pos_label=0).*

Best Random Forest hyperparameters (from GridSearchCV): max_depth=None, min_samples_leaf=1, min_samples_split=2, n_estimators=100. **Best CV malignant-F1**: 0.9506 (mean = 0.9506, std = 0.0230 across 5 folds). Notably, those values coincide with the scikit-learn defaults, which is why the “optimized” Random Forest reproduces the untuned Random Forest's test-set metrics exactly — **tuning did not improve test-set performance in this run**, and this is reported honestly rather than hidden.

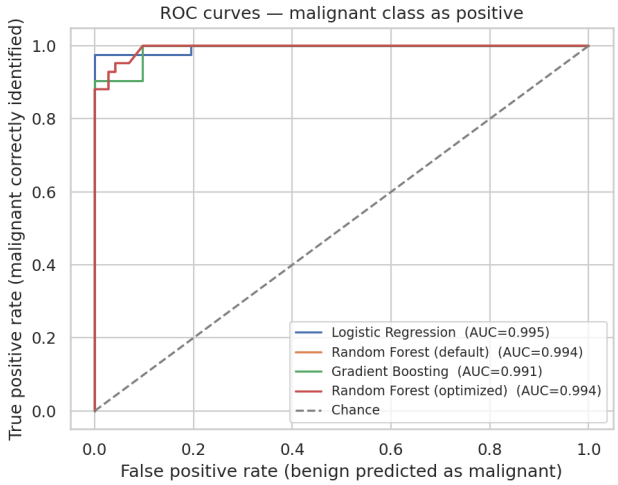


Figure 1. ROC curves for all four models with malignant as the positive class (held-out test set, n=114).

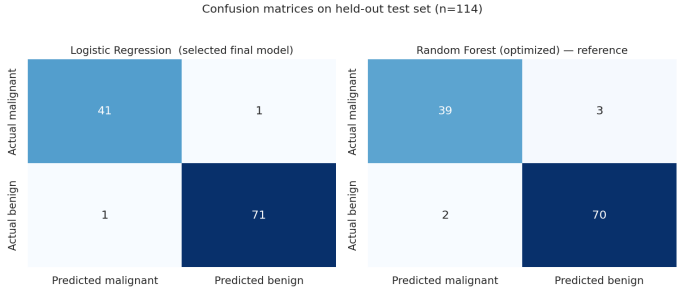


Figure 2. Confusion matrices on the held-out test set. Left: final selected model (Logistic Regression) — 1 false negative on 42 malignant cases. Right: Optimized Random Forest for reference — 3 false negatives on the same test set.

6. Final Model Selection

The primary selection criterion is **malignant F1 on the held-out test set**, with ROC-AUC and malignant recall as tie-breakers, and training time and interpretability as secondary factors. On this evidence, **Logistic Regression is selected as the final model**. It achieves the highest malignant F1 (**0.9762**), the highest ROC-AUC (**0.9954**), the highest malignant recall (**0.9762**) and the shortest fit time by two orders of magnitude. Random Forest and Gradient Boosting both land at 95.6 % accuracy but miss more malignant cases (recall 0.9286 and 0.9048 respectively).

This outcome — a linear baseline beating two ensembles — is a legitimate and well-known characteristic of the Breast Cancer Wisconsin dataset: its 30 engineered features are already highly discriminative and nearly linearly separable after standardisation. On only 455 training samples, the extra variance capacity of a tree ensemble does not translate into better generalisation. GridSearchCV further confirmed that no Random Forest configuration in the search beat the defaults. Had the dataset been larger or less linearly separable, the ensemble methods would plausibly have won.

Clinical trade-off. Even the chosen model produces 1 false negative on 42 malignant test cases. A real screening system would be expected to drive malignant recall still higher — most likely by lowering the decision threshold at the

cost of more false positives — and would require far more validation than this experiment provides. This project is educational, not clinical.

7. Limitations and Conclusion

Limitations. The test set contains only 114 observations, so single misclassifications swing recall by ~2.4 pp; conclusions from differences this small should be made with care. Only one random seed and one split were used; a more rigorous study would repeat the whole pipeline with multiple seeds. Impurity-based feature importance reflects how trees split, not clinical causation, and must not be used for medical interpretation. The dataset itself (1995) does not represent current imaging equipment, protocols or demographics.

Conclusion. The pipeline implements every requested step — exploration, stratified split, baseline, two ensembles, GridSearchCV tuning, cross-validation, visualisation and evidence-based selection — and reports an honest result: on this dataset, the regularised linear baseline is the strongest model, GridSearchCV did not improve the Random Forest, and malignant recall is the metric that would matter most in any clinical extension of this work.

8. Reproducibility and Artifacts

The entire experiment is deterministic given random_state=42 for the split, each classifier and the stratified K-fold CV. A supplementary 5-fold cross-validation on the training set (independent of the GridSearchCV run) confirms that the ordering holds beyond the single test-set split: Logistic Regression leads on mean malignant F1, followed by the two Random Forests, with Gradient Boosting last.

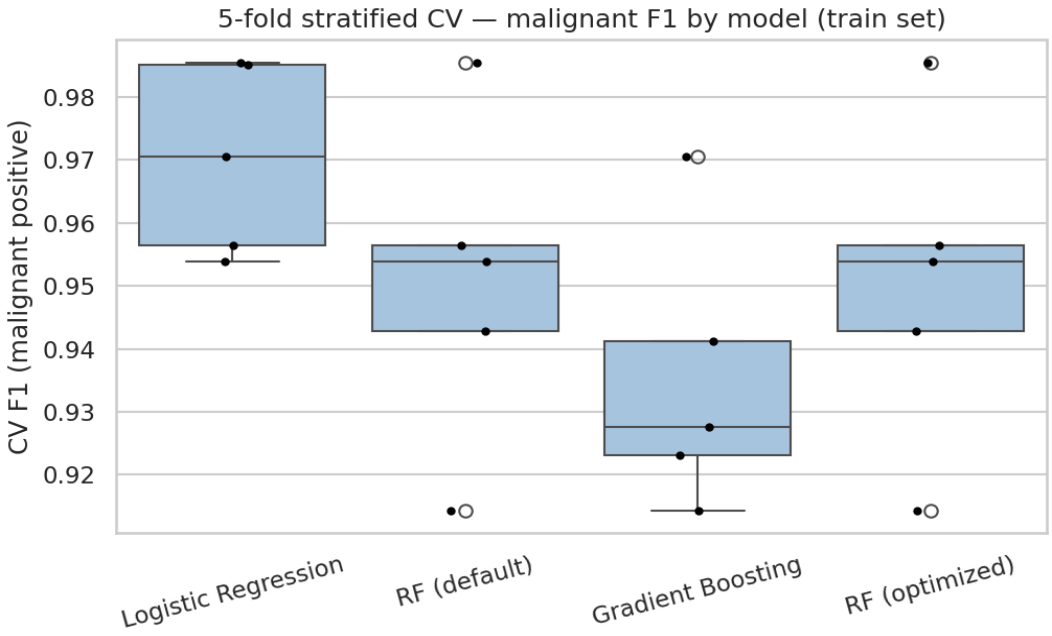


Figure 3. 5-fold stratified cross-validation on the training set (malignant-F1 scorer, independent of the GridSearchCV run). The ordering of the four models is consistent with the held-out test-set results in Table 1.

Submission folder Task_5_Ensemble_Learning/ contains: the executed Jupyter notebook (Task_5_Ensemble_Learning.ipynb), this report (Task_5_Ensemble_Learning_Report.pdf), README.md, requirements.txt, environment_info.txt, five figures under figures/, and the result exports under results/ (model_comparison.csv, random_forest_cv_results.csv, best_parameters.json and final_selection.json). The notebook was executed end-to-end from a clean kernel; all numbers in this report are drawn directly from that execution and from the exported CSV/JSON files.